WORKED SOLUTION

STEP III 1989, Question 3

Pure • Matrix Groups

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/89-s3-q3

The key idea

$\mathbf{M}$ is a rotation through $2\pi/m$, so $\mathbf{M}^m = \mathbf{I}$, and the sum of its powers is a geometric series: multiply it by $\mathbf{I} - \mathbf{M}$ and everything telescopes. For the recurrence, write out $\mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3$ and the pattern is plain. The group then comes from one identity: substituting one of these expressions into another just adds the indices, so the elements behave like the integers modulo m.

Solution

The sum of powers of $\mathbf{M}$

$\mathbf{M}$ is the matrix of an anticlockwise rotation through $\theta = 2\pi/m$ about the origin. Applying it k times rotates through $k\theta$, so

$$\mathbf{M}^k = \begin{pmatrix} \cos k\theta & -\sin k\theta \\ \sin k\theta & \cos k\theta \end{pmatrix}, \qquad \text{and in particular} \qquad \mathbf{M}^m = \begin{pmatrix} \cos 2\pi & -\sin 2\pi \\ \sin 2\pi & \cos 2\pi \end{pmatrix} = \mathbf{I}.$$

(The formula for $\mathbf{M}^k$ follows by induction from the compound-angle formulae, since $\mathbf{M}^{k+1} = \mathbf{M}\mathbf{M}^k$.)

Let $\mathbf{S} = \mathbf{I} + \mathbf{M} + \cdots + \mathbf{M}^{m-1}$. Since all these matrices are powers of $\mathbf{M}$, they commute, and the product telescopes:

$$(\mathbf{I} - \mathbf{M})\mathbf{S} = \mathbf{I} - \mathbf{M}^m = \mathbf{I} - \mathbf{I} = \mathbf{O}.$$

? **Ask yourself** Can we just "cancel" the factor $\mathbf{I} - \mathbf{M}$?

Only if $\mathbf{I} - \mathbf{M}$ is invertible: products of non-zero matrices can be $\mathbf{O}$. Here

$$\det(\mathbf{I} - \mathbf{M}) = (1 - \cos\theta)^2 + \sin^2\theta = 2 - 2\cos\theta,$$

and since $m > 1$ we have $0 < \theta \leq \pi$, so $\cos\theta \neq 1$ and the determinant is non-zero.

Multiplying $(\mathbf{I}-\mathbf{M})\mathbf{S}=\mathbf{O}$ on the left by $(\mathbf{I}-\mathbf{M})^{-1}$ gives $\mathbf{S}=\mathbf{O}$, which is the required result.

A formula for $\mathbf{X}_k$

Working out the first few terms,

$$\mathbf{X}_1=\mathbf{P}\mathbf{X}_0+\mathbf{Q},\quad \mathbf{X}_2=\mathbf{P}^2\mathbf{X}_0+\mathbf{P}\mathbf{Q}+\mathbf{Q},\quad \mathbf{X}_3=\mathbf{P}^3\mathbf{X}_0+(\mathbf{P}^2+\mathbf{P}+\mathbf{I})\mathbf{Q},$$

suggests that, for $k\geq 0$,

$$\mathbf{X}_k=\mathbf{P}^k\mathbf{X}_0+\mathbf{S}_k\mathbf{Q},\qquad \text{where}\quad \mathbf{S}_k=\mathbf{I}+\mathbf{P}+\cdots+\mathbf{P}^{k-1}\quad(\mathbf{S}_0=\mathbf{O}).$$

For $k=0$ this reads $\mathbf{X}_0=\mathbf{X}_0$, which is true. Suppose it holds for some $k\geq 0$. Then

$$\begin{aligned}\mathbf{X}_{k+1}=\mathbf{P}\mathbf{X}_k+\mathbf{Q}&=\mathbf{P}^{k+1}\mathbf{X}_0+(\mathbf{P}+\mathbf{P}^2+\cdots+\mathbf{P}^k)\mathbf{Q}+\mathbf{Q}\\&=\mathbf{P}^{k+1}\mathbf{X}_0+\mathbf{S}_{k+1}\mathbf{Q},\end{aligned}$$

so it holds for $k+1$, and by induction for all $k\geq 0$. Keep the order of the factors: $\mathbf{P}$ multiplies on the left throughout, and $\mathbf{P}$ and $\mathbf{Q}$ need not commute.

The group

Think of $\mathbf{X}_i$ as the expression $\mathbf{P}^i\mathbf{X}_0+\mathbf{S}_i\mathbf{Q}$, a rule that turns any matrix $\mathbf{X}_0$ into another. Substituting $\mathbf{X}_j$ for $\mathbf{X}_0$ gives

$$\begin{aligned}\mathbf{X}_i*\mathbf{X}_j&=\mathbf{P}^i(\mathbf{P}^j\mathbf{X}_0+\mathbf{S}_j\mathbf{Q})+\mathbf{S}_i\mathbf{Q}\\&=\mathbf{P}^{i+j}\mathbf{X}_0+(\mathbf{P}^i\mathbf{S}_j+\mathbf{S}_i)\mathbf{Q}.\end{aligned}$$

Now $\mathbf{P}^i\mathbf{S}_j=\mathbf{P}^i+\cdots+\mathbf{P}^{i+j-1}$, and adding $\mathbf{S}_i=\mathbf{I}+\cdots+\mathbf{P}^{i-1}$ gives exactly $\mathbf{S}_{i+j}$. So, for any $\mathbf{P}$,

$$\mathbf{X}_i*\mathbf{X}_j=\mathbf{X}_{i+j}.$$

(This is no surprise: doing j steps of the recurrence and then i more is doing $i+j$ steps.)

Now take $\mathbf{P}=\mathbf{M}$. By the first part, $\mathbf{M}^m=\mathbf{I}$ and $\mathbf{S}_m=\mathbf{O}$, so

$$\mathbf{X}_m=\mathbf{X}_0,$$

that is, $\mathbf{X}_m$ is the expression that leaves every matrix unchanged. Then for any k, $\mathbf{X}_{k+m}=\mathbf{X}_k*\mathbf{X}_m=\mathbf{X}_k$, so the sequence of expressions repeats with period m and

$$G=\{\mathbf{X}_1,\mathbf{X}_2,\mathbf{X}_3,\ldots\}=\{\mathbf{X}_1,\mathbf{X}_2,\ldots,\mathbf{X}_m\}$$

is a finite set. We check the group axioms.

- *Closure.* $\mathbf{X}_i*\mathbf{X}_j=\mathbf{X}_{i+j}$, and $i+j\geq 1$, so this lies in G.
- *Associativity.* Both $(\mathbf{X}_i*\mathbf{X}_j)*\mathbf{X}_k$ and $\mathbf{X}_i*(\mathbf{X}_j*\mathbf{X}_k)$ equal $\mathbf{X}_{i+j+k}$. (More generally, substitution is composition of functions, which is always associative.)

- *Identity.* $\mathbf{X}_m * \mathbf{X}_j = \mathbf{X}_{m+j} = \mathbf{X}_j$ and $\mathbf{X}_j * \mathbf{X}_m = \mathbf{X}_j$, so $\mathbf{X}_m$ is the identity.
- *Inverses.* For $1 \le i < m$, $\mathbf{X}_i * \mathbf{X}_{m-i} = \mathbf{X}_{m-i} * \mathbf{X}_i = \mathbf{X}_m$, and $\mathbf{X}_m$ is its own inverse. Each inverse lies in G.

So G is a finite group under $*$.

In fact it has exactly m elements: $\mathbf{X}_1, \ldots, \mathbf{X}_m$ are different expressions because their coefficients $\mathbf{M}^1, \ldots, \mathbf{M}^m$ are rotations through different angles in $(0, 2\pi]$. Since every element is a "power" of $\mathbf{X}_1$, G is the cyclic group of order m.

Checks

- With $m = 2$, $\mathbf{M} = -\mathbf{I}$ and the sum is $\mathbf{I} + (-\mathbf{I}) = \mathbf{O}$. With $m = 4$, $\mathbf{M}$ is a quarter-turn and $\mathbf{I} + \mathbf{M} + \mathbf{M}^2 + \mathbf{M}^3 = \mathbf{I} + \mathbf{M} - \mathbf{I} - \mathbf{M} = \mathbf{O}$.
- Numerically, the sum of powers is $\mathbf{O}$ (to rounding) for every m from 2 to 39; the formula for $\mathbf{X}_k$ matches the recurrence for random $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{X}_0$; and with $\mathbf{P} = \mathbf{M}$, random $\mathbf{Q}$ and $\mathbf{X}_0$, running the recurrence m steps returns exactly to $\mathbf{X}_0$.

Common slips

- Cancelling $\mathbf{I} - \mathbf{M}$ without checking it is invertible.
- Writing $\mathbf{Q}\mathbf{S}_k$ or $\mathbf{X}_0\mathbf{P}^k$: matrices do not commute, and the recurrence puts $\mathbf{P}$ on the left.
- Taking $\mathbf{X}_0$ as the identity: $\mathbf{X}_0$ is not in G as listed; the identity element is $\mathbf{X}_m$, which equals $\mathbf{X}_0$ only because $\mathbf{S}_m = \mathbf{O}$.

Beyond STEP

Each $\mathbf{X}_i$ is an *affine map* $\mathbf{X} \mapsto \mathbf{A}\mathbf{X} + \mathbf{B}$, and the affine maps with invertible $\mathbf{A}$ form a group under composition; here we found a cyclic subgroup of order m generated by a single map whose mth power is the identity. The fact that $\mathbf{I} + \mathbf{M} + \cdots + \mathbf{M}^{m-1} = \mathbf{O}$ is the matrix version of the mth roots of unity summing to zero, and the general idea, a group acting on a set by transformations, is central to a first course in group theory. Allenby's *Rings, Fields and Groups* develops this.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.